OPTIMAL LINEAR FILTERING IN MEAN SQUARE SENSE: THE WIENER FILTER

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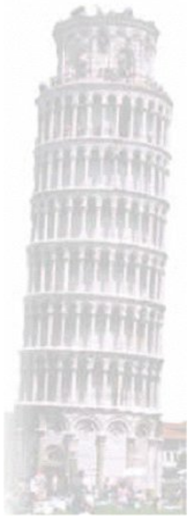


Optimal Linear Filtering

- Let us consider some important **applications** of the **LMMSE estimate**.

LMMSE estimator $\rightarrow$
$$\begin{cases} \hat{\boldsymbol{\theta}}_{LMMSE} = \boldsymbol{\eta}_\theta + \mathbf{C}_{X\theta}^T \mathbf{C}_X^{-1} \left(\mathbf{X} - \boldsymbol{\eta}_X \right) \\ MSE\left\{ \hat{\boldsymbol{\theta}}_{LMMSE} \right\} = \mathbf{C}_\varepsilon = \mathbf{C}_\theta - \mathbf{C}_{X\theta}^T \mathbf{C}_X^{-1} \mathbf{C}_{X\theta} \end{cases}$$

Assumptions:

- **A1)** Observed data = SOI + Noise, signal and noise are uncorrelated:

$X[n]=S[n]+W[n]$, $n=0,1,2, \ldots, N\text{-}1 \rightarrow \mathbf{X}=\mathbf{S}+\mathbf{W}$

- **A2)** The signal we want to estimate is a realization of a WSS process, not necessarily Gaussian, with zero mean and known autocorrelation function (ACF), i.e. known power spectral density (PSD).

- **A3)** The noise is additive, not necessarily Gaussian, not necessarily white, with zero mean and known ACF, i.e. known PSD.

Optimal Linear Filtering

- A2) and A3) imply that $\boldsymbol{\eta}_\theta = 0$ and $\boldsymbol{\eta}_X = 0$
- A1) implies that:

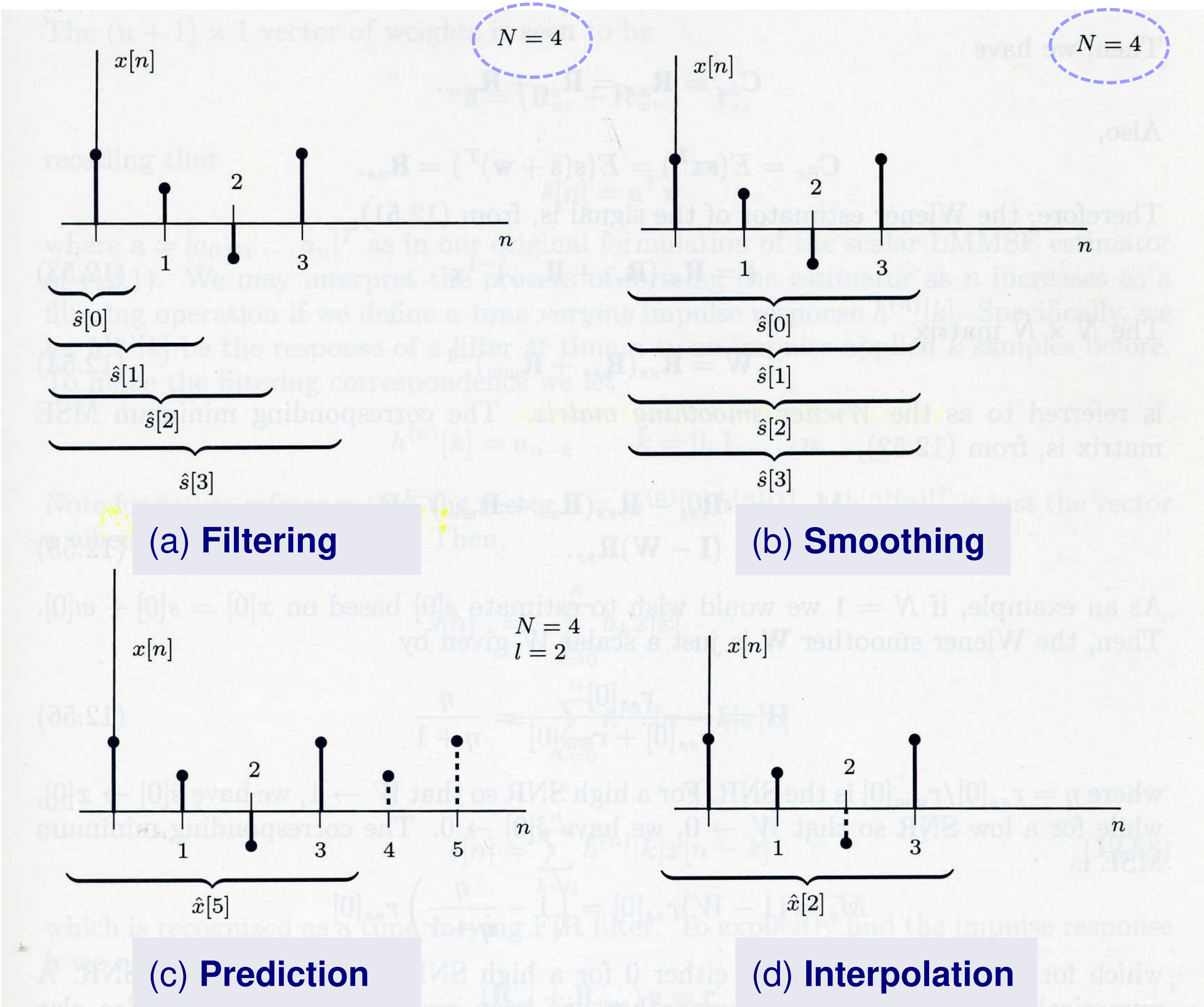
$$\mathbf{C}_X = E\left\{\mathbf{X}\mathbf{X}^T\right\} = E\left\{\mathbf{S}\mathbf{S}^T\right\} + E\left\{\mathbf{W}\mathbf{W}^T\right\} = \mathbf{C}_S + \mathbf{C}_W$$

$$\hat{\boldsymbol{\theta}}_{LMMSE} = \mathbf{C}_{X\theta}^T \left(\mathbf{C}_S + \mathbf{C}_W\right)^{-1} \mathbf{X} = \mathbf{A}\mathbf{X} \quad \Rightarrow \quad \mathbf{A} \equiv \text{Wiener Smoothing Matrix}$$

- There are various kind of problems that can be solved with the theory of the **LMMSE estimator** $\rightarrow$ **Wiener Filter**
- We have 4 different kinds of **Wiener Filter**, according to what is the vector parameter $\boldsymbol{\theta}$ to be estimated and what are the observed data we use to estimate it:

(1) **Filtering**, (2) **Smoothing**, (3) **Prediction**, (4) **Interpolation**.

Optimal Linear Filtering



(a) **Filtering**

(b) **Smoothing**

(c) **Prediction**

(d) **Interpolation**

Smoothing Wiener Filter (Batch)

Smoothing :

$$\begin{cases} \boldsymbol{\theta} = \mathbf{S} = \left[S[0] \quad S[1] \quad \cdots \quad S[N-1] \right]^T \\ \mathbf{X} = \left[X[0] \quad X[1] \quad \cdots \quad X[N-1] \right]^T = \mathbf{S} + \mathbf{W} \\ \mathbf{C}_{X\theta} = \mathbf{C}_{XS} = E\left\{ \mathbf{X}\mathbf{S}^T \right\} = E\left\{ (\mathbf{S}+\mathbf{W})\mathbf{S}^T \right\} = E\left\{ \mathbf{S}\mathbf{S}^T \right\} = \mathbf{C}_S \end{cases}$$

$$\hat{\mathbf{S}}_{LMMSE} = \mathbf{C}_S(\mathbf{C}_S + \mathbf{C}_W)^{-1}\mathbf{X} = \mathbf{A}\mathbf{X} \quad \Rightarrow \quad \mathbf{A} \equiv \text{Wiener Smoothing Matrix}$$

$$\mathbf{C}_\varepsilon = \mathbf{C}_S - \mathbf{C}_S(\mathbf{C}_S + \mathbf{C}_W)^{-1}\mathbf{C}_S = \left(\mathbf{I} - \mathbf{C}_S(\mathbf{C}_S + \mathbf{C}_W)^{-1} \right)\mathbf{C}_S = (\mathbf{I} - \mathbf{A})\mathbf{C}_S$$

- The **smoothing Wiener filter** working in **batch mode** is a **linear time-variant (LTV)** filter, as we will see shortly.

Smoothing Wiener Filter (Batch)

- The **smoothing Wiener filter** operating in **batch mode** is a **linear time-variant (LTV) filter**, since in general the data samples $\{X[n]\}$'s are processed with a linear filter whose impulse response changes with discrete-time n.

- The case of **constant signal** is an exception, due to the fact that the signal samples are fully correlated, i.e. $S[n]=S$ for any n, hence the correlation between $S[n]$ and the data vector $\mathbf{X}$ does not change with time n.

- In fact, when the problem is to estimate a constant signal $S[n]=S$ in additive white noise, we have:

$$\mathbf{C}_X = \mathbf{C}_S + \sigma_W^2 \mathbf{I}$$

$$C_S[m] = E\left\{\left(S[n]-\eta_S\right)\left(S[n+m]-\eta_S\right)\right\} = E\left\{\left(S-\eta_S\right)^2\right\} = \sigma_S^2 \quad \forall m$$

$$\left[\mathbf{C}_S\right]_{i+1,k+1} = \sigma_S^2, \quad i,k = 0,1,2,\cdots,N-1 \quad \Rightarrow \quad \mathbf{C}_S = \sigma_S^2 \mathbf{1}\mathbf{1}^T$$

Smoothing Wiener Filter (Batch)

- Alternatively, the signal covariance matrix can be found as follows:

$$\mathbf{C}_S = E\left\{(\mathbf{S}-\boldsymbol{\eta}_S)(\mathbf{S}-\boldsymbol{\eta}_S)^T\right\} = E\left\{(S\mathbf{1}-\eta_S\mathbf{1})(S\mathbf{1}-\eta_S\mathbf{1})^T\right\}$$

$$= E\left\{(S-\eta_S)^2\mathbf{1}\mathbf{1}^T\right\} = E\left\{(S-\eta_S)^2\right\}\mathbf{1}\mathbf{1}^T = \sigma_S^2\mathbf{1}\mathbf{1}^T$$

- Hence, the observed data matrix and its inverse are as follows:

$$\mathbf{C}_X = \mathbf{C}_S + \mathbf{C}_W = \sigma_S^2\mathbf{1}\mathbf{1}^T + \sigma_W^2\mathbf{I} = \sigma_W^2\left(\gamma\mathbf{1}\mathbf{1}^T + \mathbf{I}\right)$$

where $\gamma \triangleq \dfrac{\sigma_S^2}{\sigma_W^2}$ Signal-to-Noise power Ratio (SNR)

$$\mathbf{C}_X^{-1} = \frac{1}{\sigma_W^2}\left(\gamma\mathbf{1}\mathbf{1}^T + \mathbf{I}\right)^{-1} = \frac{1}{\sigma_W^2}\left(\mathbf{I} - \frac{\gamma}{1+\gamma\mathbf{1}^T\mathbf{1}}\mathbf{1}\mathbf{1}^T\right)$$

Woodbury Identity

Smoothing Wiener Filter (Batch)

$$\hat{\mathbf{S}}_{LMMSE} = \mathbf{C}_S(\mathbf{C}_S + \mathbf{C}_W)^{-1}\mathbf{X} = \mathbf{A}\mathbf{X} \quad (\eta_S = 0) \qquad \mathbf{A} \equiv \text{Wiener Smoothing Matrix}$$

$$\hat{\mathbf{S}}_{LMMSE} = \mathbf{C}_S(\mathbf{C}_S + \mathbf{C}_W)^{-1}\mathbf{X} = \sigma_S^2\mathbf{1}\mathbf{1}^T\left(\sigma_S^2\mathbf{1}\mathbf{1}^T + \sigma_W^2\mathbf{I}\right)^{-1}\mathbf{X}$$

$$= \sigma_S^2\mathbf{1}\mathbf{1}^T \cdot \frac{1}{\sigma_W^2}\left(\mathbf{I} - \frac{\gamma}{1+\gamma\mathbf{1}^T\mathbf{1}}\mathbf{1}\mathbf{1}^T\right)\mathbf{X} = \gamma\mathbf{1}\mathbf{1}^T\left(\mathbf{I} - \frac{\gamma}{1+\gamma\mathbf{1}^T\mathbf{1}}\mathbf{1}\mathbf{1}^T\right)\mathbf{X}$$

$$= \gamma\mathbf{1}\left(\mathbf{1}^T - \frac{\gamma}{1+N\gamma}\mathbf{1}^T\mathbf{1}\mathbf{1}^T\right)\mathbf{X} = \gamma\mathbf{1}\left(1 - \frac{N\gamma}{1+N\gamma}\right)\mathbf{1}^T\mathbf{X}$$

$$= \mathbf{1}\frac{N\gamma}{1+N\gamma}\cdot\frac{1}{N}\mathbf{1}^T\mathbf{X} = \left(\alpha\bar{\mathbf{X}}\right)\mathbf{1} = \hat{S}_{LMMSE}\mathbf{1}$$

where $\alpha \triangleq \dfrac{N\gamma}{1+N\gamma}$, $\bar{\mathbf{X}}$ is the Sample Mean, and $\hat{S}_{LMMSE} = \alpha\bar{\mathbf{X}}$

- that is in agreement with what we have already found for η_s=0.

Smoothing Wiener Filter (Batch)

- The **Wiener smoothing matrix** in this particular case of constant signal is:

$$\mathbf{A} = \mathbf{C}_S(\mathbf{C}_S + \mathbf{C}_W)^{-1} = \mathbf{1}\frac{N\gamma}{1+N\gamma}\cdot\frac{1}{N}\mathbf{1}^T = \frac{\alpha}{N}\mathbf{1}\mathbf{1}^T = \frac{\alpha}{N}\underset{N\times N}{\begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{bmatrix}}$$

$$\underset{N\times 1}{\hat{\mathbf{S}}_{LMMSE}} = \begin{bmatrix} \hat{S}[0] \\ \hat{S}[1] \\ \vdots \\ \hat{S}[N-1] \end{bmatrix} = \frac{\alpha}{N}\begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{bmatrix}\cdot\begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix} = \begin{bmatrix} \hat{S}_{LMMSE} \\ \hat{S}_{LMMSE} \\ \vdots \\ \hat{S}_{LMMSE} \end{bmatrix} = \hat{S}_{LMMSE}\mathbf{1}$$

- Hence, in the case of completely correlated signal $S[n]=S$ embedded in white noise $W[n]$, all the data $\{X[n]\}$ are weighted in the same way, whatever is the discrete-time n, since all the rows of the matrix **A** are equal.

Smoothing Wiener Filter (Batch)

- In the more general case of **partially correlated signal**, the rows of the Wiener smoothing matrix are no more all equal and hence the matrix represents a **linear time-variant filter (LTV)**:

$$\underset{N\times 1}{\hat{\mathbf{S}}_{LMMSE}} = \begin{bmatrix} \hat{S}[0] \\ \hat{S}[1] \\ \vdots \\ \hat{S}[N-1] \end{bmatrix} = \underset{N\times N}{\mathbf{A}} \cdot \underset{N\times 1}{\mathbf{X}} = \begin{bmatrix} \mathbf{a}_0 \\ \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_{N-1} \end{bmatrix} \begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix}$$

where $\mathbf{a}_n$ is the $(n+1)$-th row of matrix $\mathbf{A}$, i.e.

$$\underset{1\times N}{\mathbf{a}_n} \triangleq \begin{bmatrix} a_{n,0} & a_{n,1} & \cdots & a_{n,N-1} \end{bmatrix}, \quad n = 0,1,2,\cdots,N-1$$

Smoothing Wiener Filter (Batch)

$$\hat{S}[n] = \mathbf{a}_n \mathbf{X} = \begin{bmatrix} a_{n,0} & a_{n,1} & \cdots & a_{n,N-1} \end{bmatrix} \begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix}$$

$$= \sum_{k=0}^{N-1} a_{n,k} X[k] = \sum_{k=0}^{N-1} h[k,n] X[k] \neq \sum_{k=0}^{N-1} h[n-k] X[k], \quad n = 0,1,2,\cdots,N-1$$

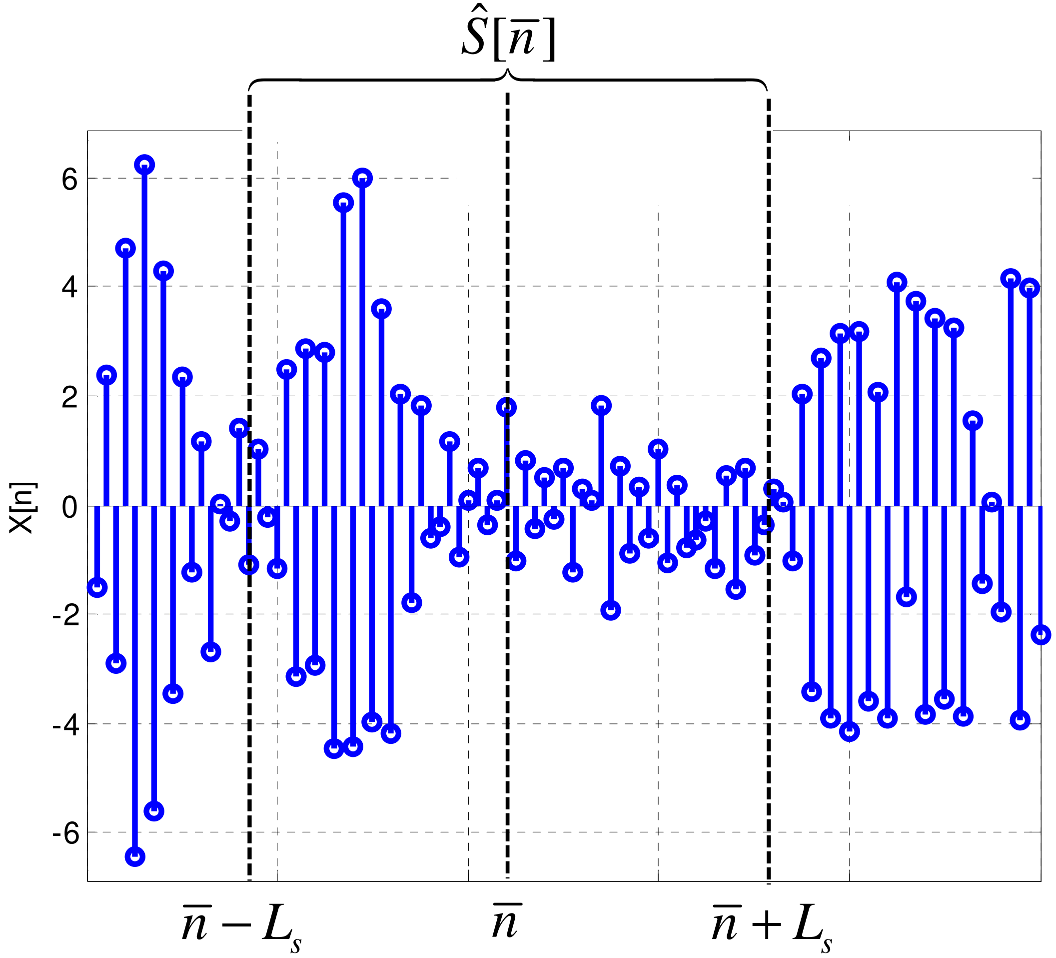
where $h[k,n] \triangleq a_{n,k}$ is the impulse response at discrete-time n

- In general the rows of matrix **A** are not equal. This implies that the impulse response of the **smoothing Wiener filter** operating in **batch mode** changes with the discrete-time *n*, i.e. it is time-variant.

- This is due to the fact that the correlation of the signal sample $S[n]$ with the data vector **X** changes with time *n*. This is true in particular for the first L_S samples and the last L_S samples of signal vector **S**, where L_S is the correlation length of process $S[n]$.

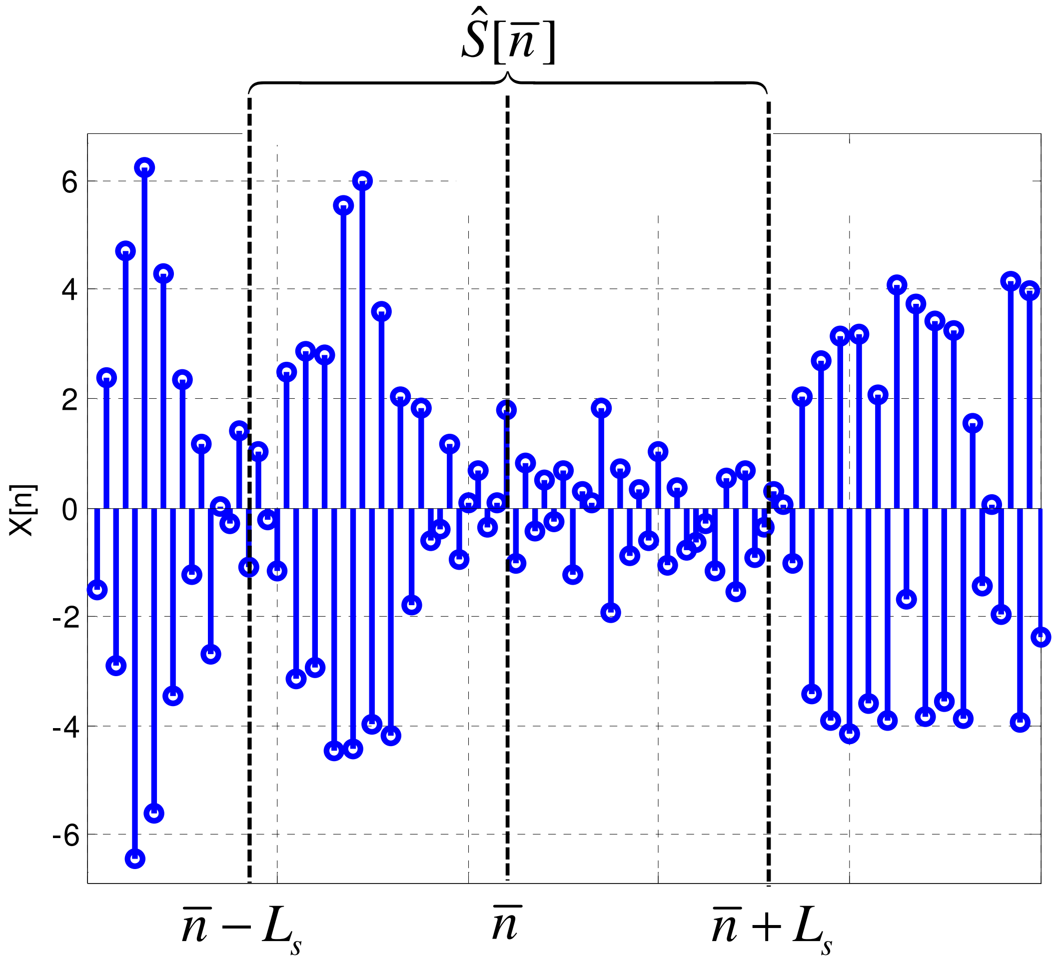
Smoothing Wiener Filter (Batch)

$$\hat{S}[\bar{n}] = \sum_{k=0}^{N-1} a_{\bar{n},k} X[k] \simeq \sum_{k=\bar{n}-L_s}^{\bar{n}+L_s} a_{\bar{n},k} X[k]$$



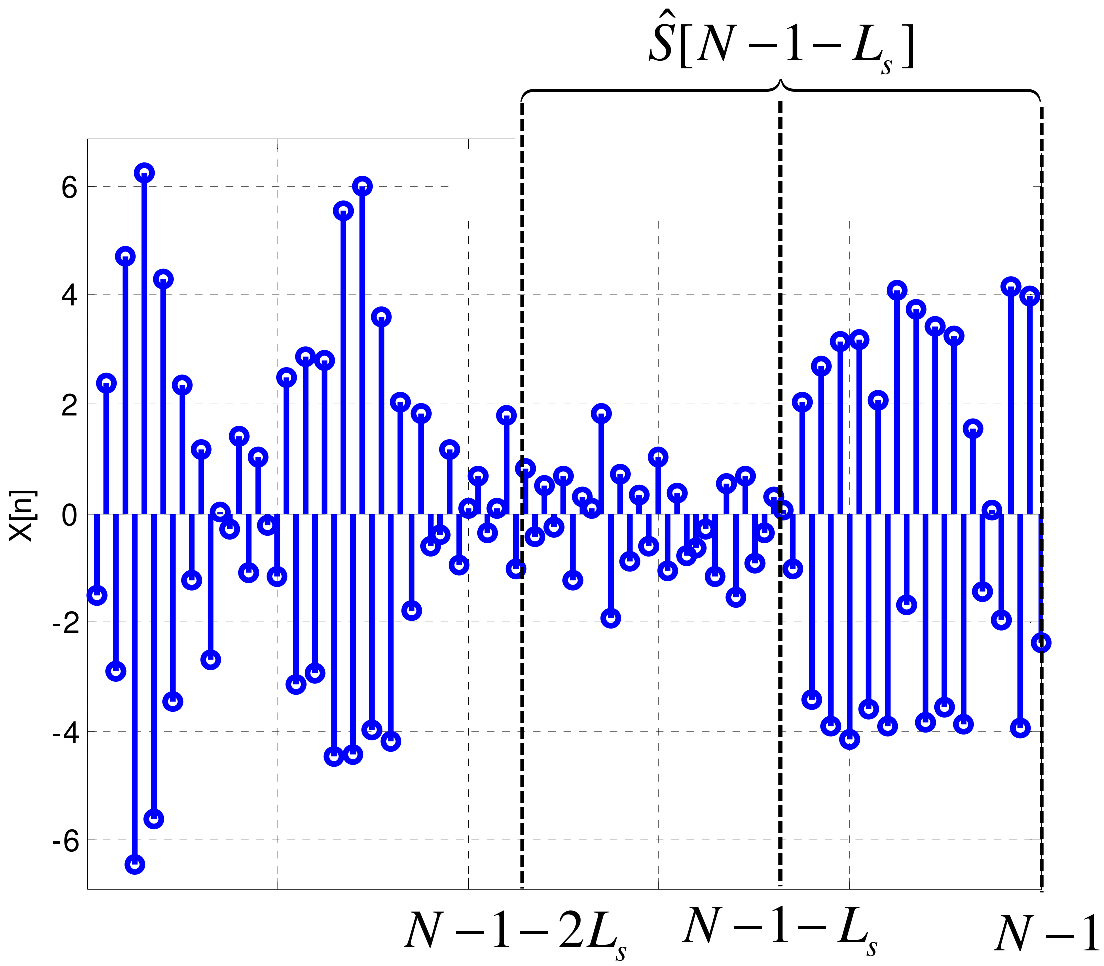
Smoothing Wiener Filter (Batch)

$MSE = \text{const.}$ for $\overline{n}$ such that: $\overline{n} - L_s \geq 0$ and $\overline{n} + L_s \leq N-1 \Rightarrow L_s \leq \overline{n} \leq N-1-L_s$



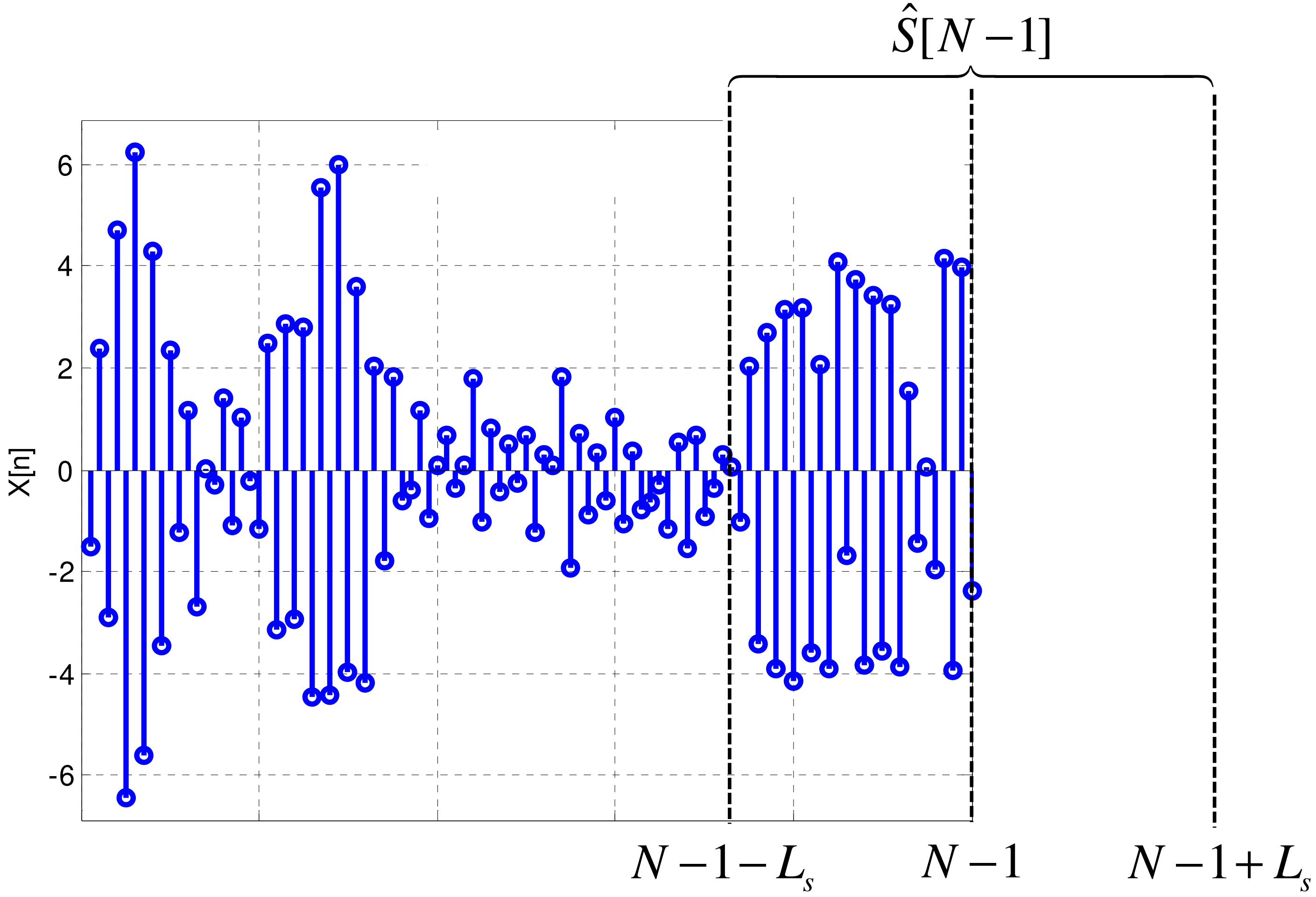
Smoothing Wiener Filter (Batch)

$$MSE = \text{const.}\ \ \text{for } \bar{n}\ \text{ such that: } \bar{n} - L_s \geq 0\ \ \text{and}\ \ \bar{n} + L_s \leq N-1\ \Rightarrow\ L_s \leq \bar{n} \leq N-1-L_s$$



Smoothing Wiener Filter (Batch)

For $\bar{n} > N-1-L_s$ the MSE increases with $\bar{n}$. The situation is symmetric for $\bar{n} < L_s$.



Smoothing Wiener Filter (Batch)

- **Numerical example:** $S[n]$ is an AR(1) process (a.k.a. Markov process)

$\sigma_S^2 = 1$ power of $S[n]$

$\rho = 0.95$ one-lag correlation coefficient of $S[n]$

$\gamma_{dB} = 10 Log_{10}(\gamma) = -10\, dB,$ where $\gamma \triangleq \dfrac{\sigma_S^2}{\sigma_W^2}$ Signal-to-Noise power Ratio (SNR)

$$I[n] \Longrightarrow \boxed{l[n]} \Longrightarrow S[n] = \rho\, S[n-1] + I[n]$$

Innovation process (white process)

IIR filter of order 1

AR(1): recursive equation of order 1

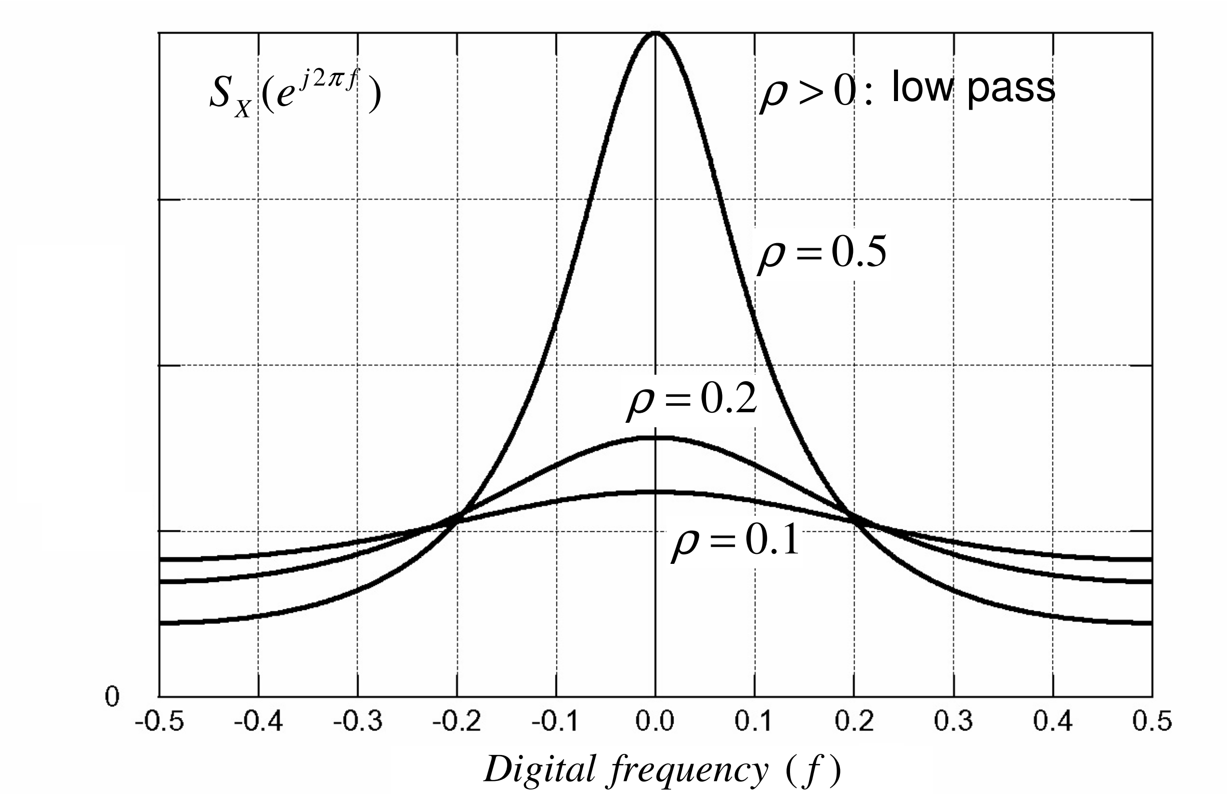
$$l[n] = \rho^n u[n] \overset{FT}{\Leftrightarrow} L\left(e^{j2\pi f}\right) = \frac{1}{1-\rho e^{-j2\pi f}}$$

Smoothing Wiener Filter (Batch)

$$R_S[m] = \sigma_S^2 \rho^{|m|}$$

ACF and PSD of an AR(1) process

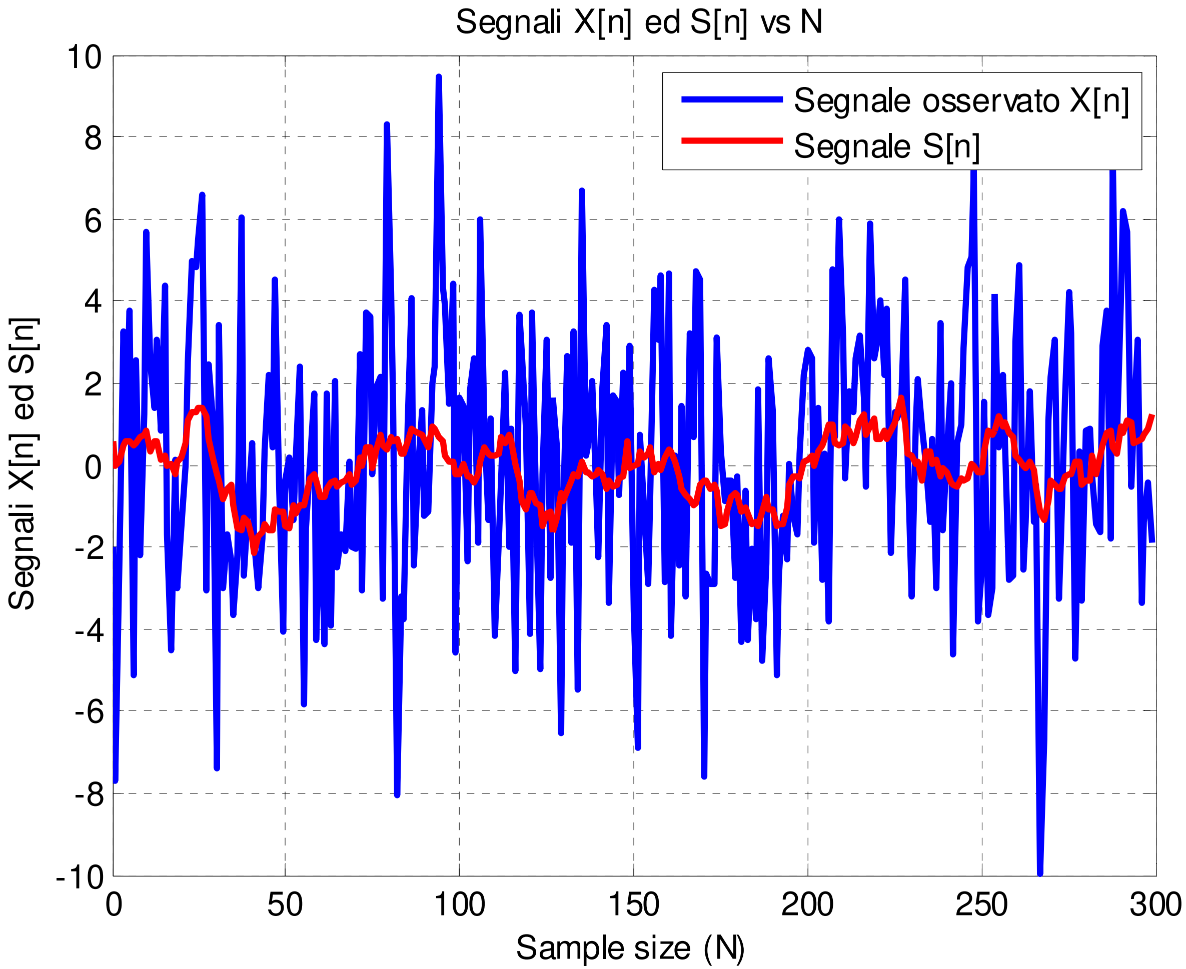
$$S_S(e^{j2\pi f}) = \sigma_I^2 \left| \frac{1}{1-\rho e^{-j2\pi f}} \right|^2 = \frac{\sigma_S^2 (1-\rho^2)}{1+\rho^2 - 2\rho \cos(2\pi f)}$$



$$\sigma_S^2 = \frac{\sigma_I^2}{1-\rho^2}$$

Smoothing Wiener Filter (Batch)

$$\gamma_{dB} = -10dB, \quad \rho = 0.95$$



Smoothing Wiener Filter (Batch)

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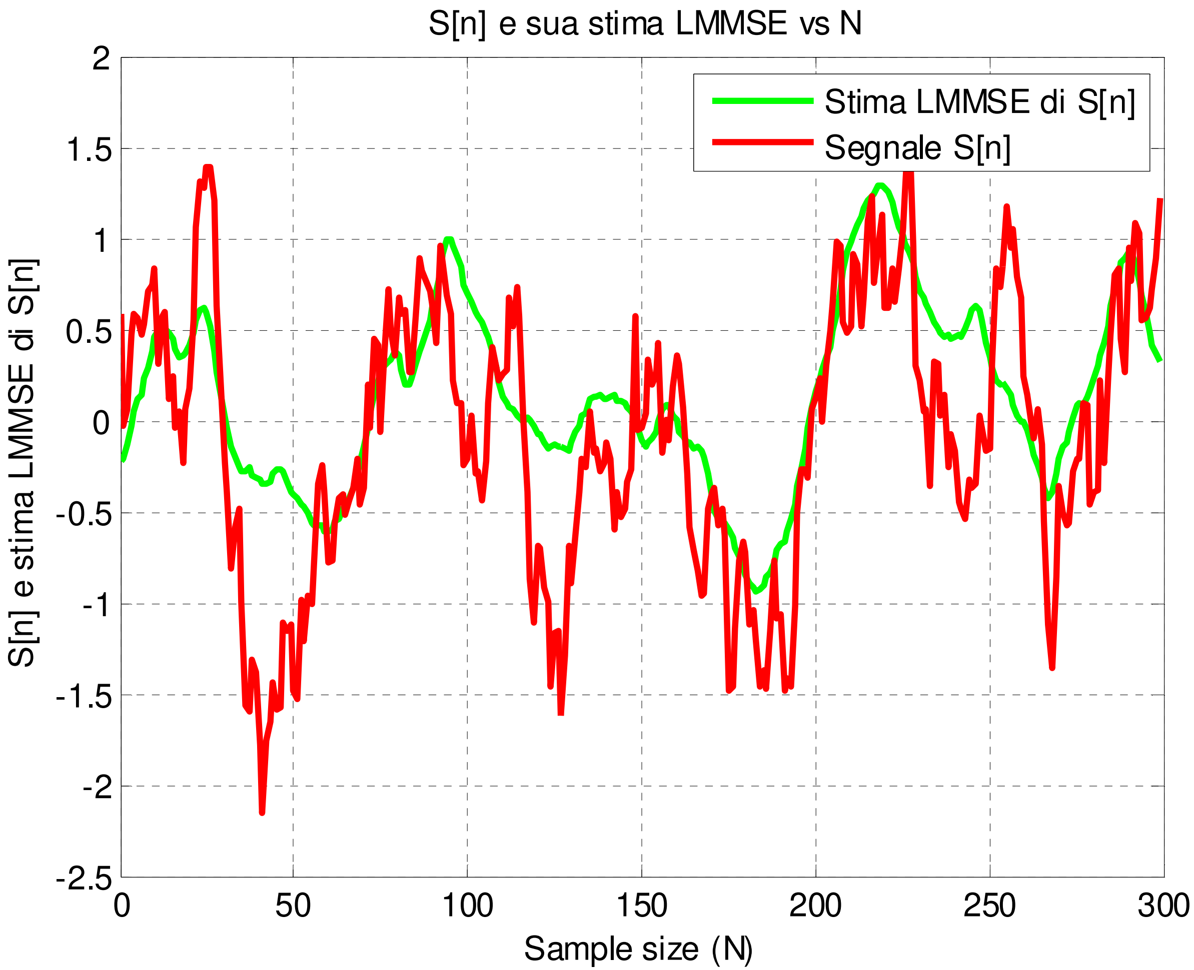

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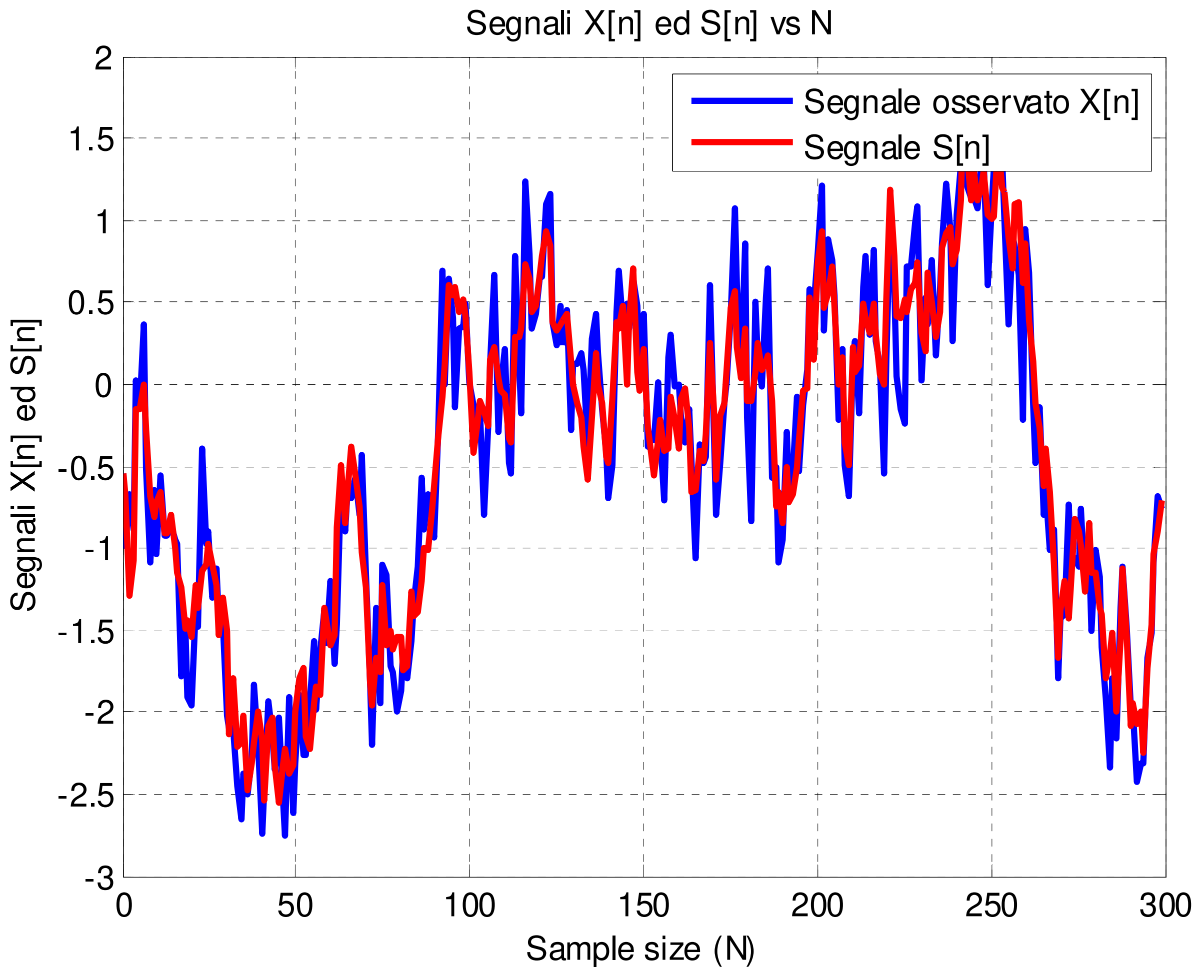

Smoothing Wiener Filter (Batch)

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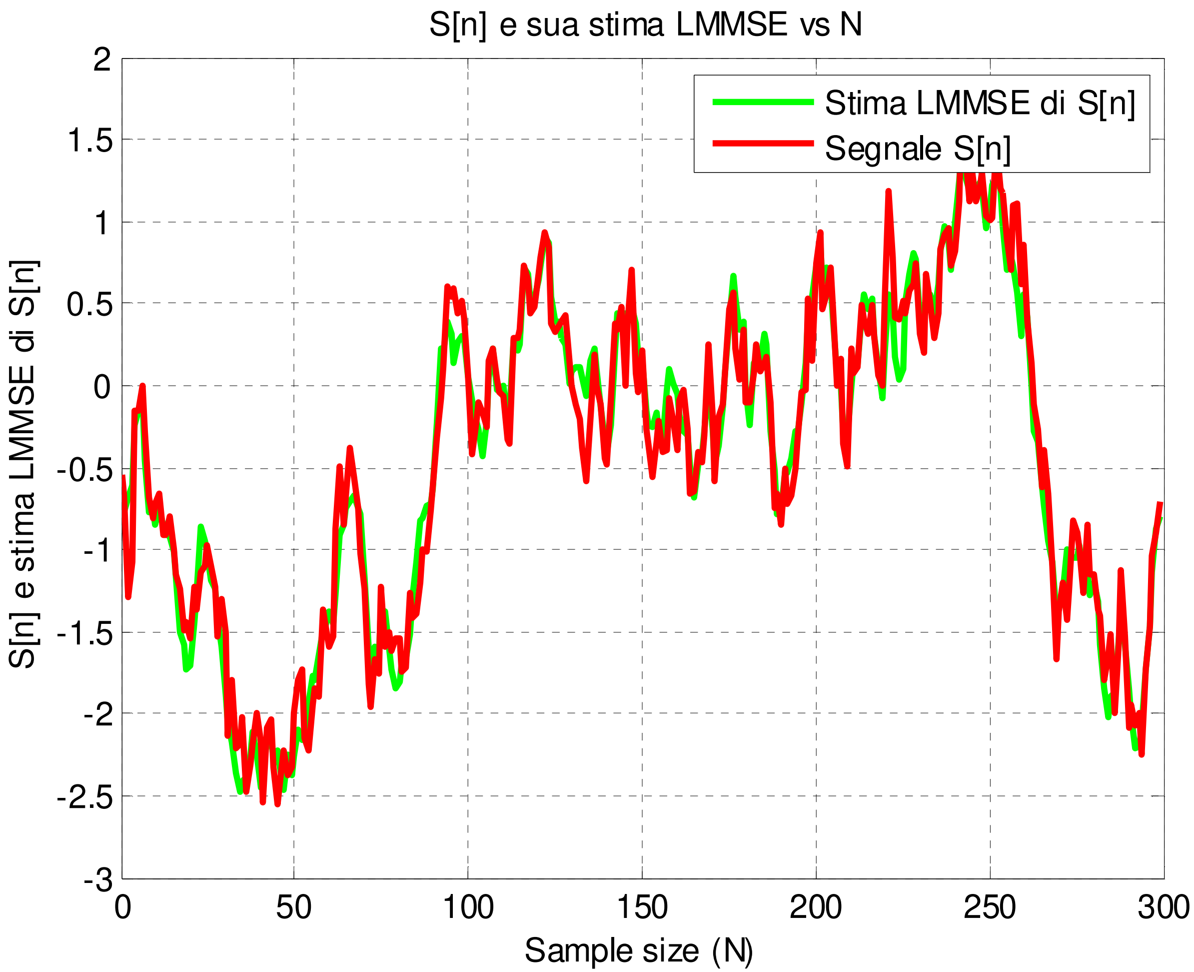
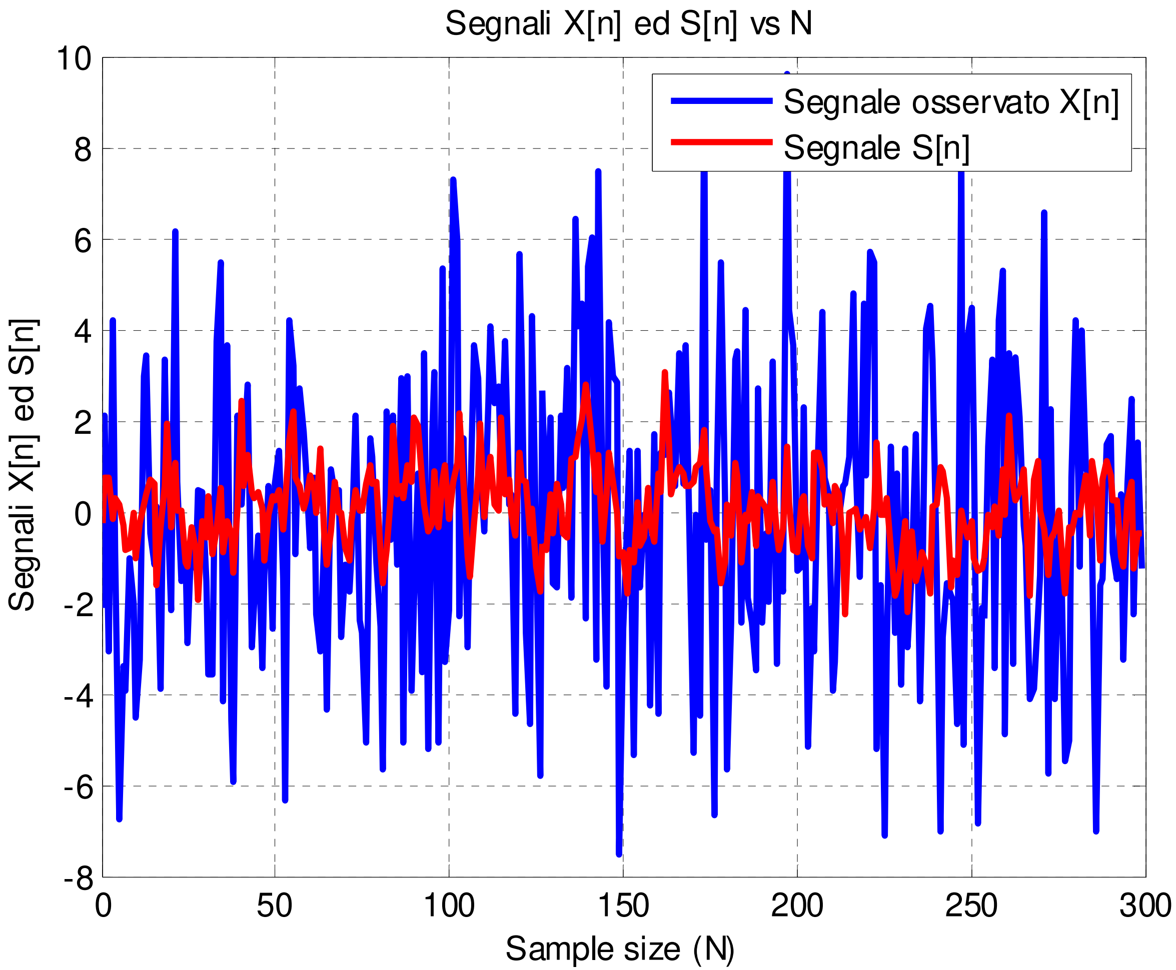


Smoothing Wiener Filter (Batch)

$$\gamma_{dB} = -10dB, \quad \rho = 0.5$$



Smoothing Wiener Filter (Batch)

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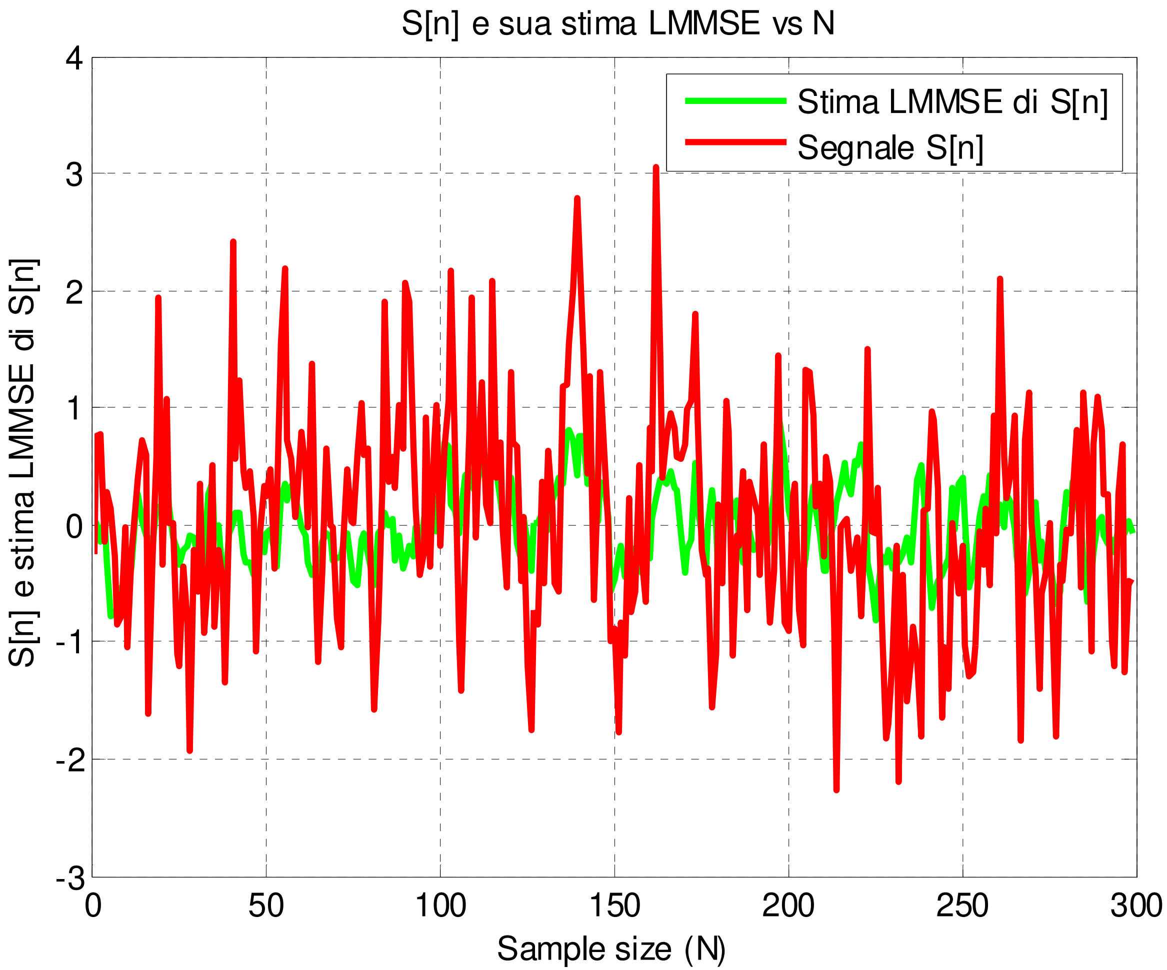
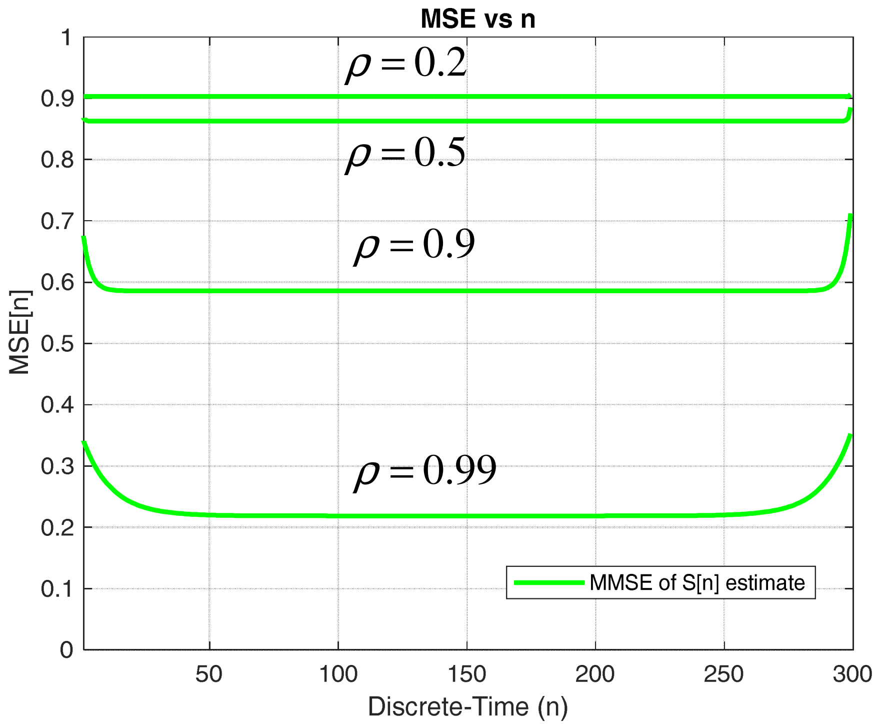


Smoothing Wiener Filter (Batch)



$$MSE\left\{\hat{S}_{LMMSE}[n]\right\}=\left[\mathbf{C}_{\varepsilon}\right]_{n+1,n+1}$$

$$N=300$$

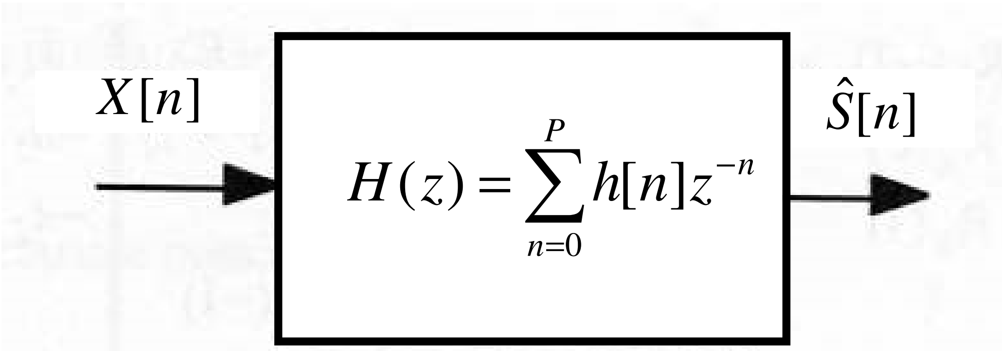
$$\gamma_{dB}=-10dB$$

- Denote by L_s is the correlation time of the process $S[n]$.
- The estimation accuracy of the first L_s and of the last L_s signal samples (i.e. at the beginning and at the end of the observation interval) is lower than for the other samples.

Wiener Filter (Sequential)

- When the signal and noise processes are WSS, we can enforce the **Wiener filter** to be not only **linear** but also **time-invariant**, i.e. **LTI**, so the estimate is made available **sequentially** at the output of the filter:

$$\hat{S}[n] = \sum_{k=0}^{P} h[k] X[n-k] = h[n] \otimes X[n], \quad n = 0,1,2,\cdots$$



$$\begin{cases} FIR: & P < +\infty \\ IIR: & P = +\infty \end{cases}$$

- We will now investigate the application of the **LTI Wiener filter** to <u>prediction</u>, <u>filtering</u>, and <u>smoothing</u>.